

Smart Home Control Hub - PIC16F877A Schematic (Text-Based Overview)

Microcontroller: PIC16F877A

Power:

- VDD -> +5V
- VSS -> GND
- MCLR -> 10k pull-up to +5V
- Crystal -> OSC1/OSC2 with 22pF caps

I2C Bus (MSSP):

- RC3 (SCL) -> DS1307 SCL, 24LC256 SCL, sensors SCL
- RC4 (SDA) -> DS1307 SDA, 24LC256 SDA, sensors SDA
- 4.7k pull-ups to +5V on SDA/SCL

UART to ESP8266:

- RC6 (TX) -> Level Shifter -> ESP8266 RX (3.3V)
- RC7 (RX) <- ESP8266 TX (3.3V safe to PIC)
- 3.3V Regulator for ESP8266

LCD 16x2 (4-bit mode):

- RD0 -> D4
- RD1 -> D5
- RD2 -> D6
- RD3 -> D7
- RD4 -> RS
- RD5 -> EN
- RW -> GND

Keypad (4x4):

- RB0..RB3 -> Columns
- RB4..RB7 -> Rows

Analog Sensors (ADC):

- RA0 (AN0) -> LM35
- RA1 (AN1) -> MQ-2
- RA2 (AN2) -> LDR divider

Digital Sensors:

- RB0 (INT) -> PIR sensor
- Additional sensors on free pins

Relay Outputs:

- RC0 -> Transistor -> Relay1
- RC1 -> Transistor -> Relay2
- Diodes across relay coils

This is a text-based schematic. For detailed drawing, a graphical tool is needed.